Retail Sales Dashboard

Power BI project | 2-page dashboard | Sales of 2025

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Objective

Build an executive dashboard that shows how a retail company is performing: revenue, profit, products, cities and customers. The dashboard lets a manager find the best and weakest areas and decide where to act.

Business questions

• How much revenue and profit did we make, and what is the margin?
• Which months, products, categories and cities perform best and worst?
• How dependent are we on a few customers?
• Do discounts hurt profit?

Key findings

• Revenue was Rs 22.58M with a profit margin of 24.0% across 240 orders. Average order value is Rs 94,079.

• November and December together bring 23% of the year's revenue; December alone grew +185% over November.

• Laptop is the biggest product (28% of revenue). Electronics is the top category with 58% of revenue.

• Stationery has the best margin (39.8%) but the smallest revenue; Electronics margin is 17.3%.

• Faisalabad is the strongest city (Rs 8.0M) and Lahore the weakest (Rs 659,778).

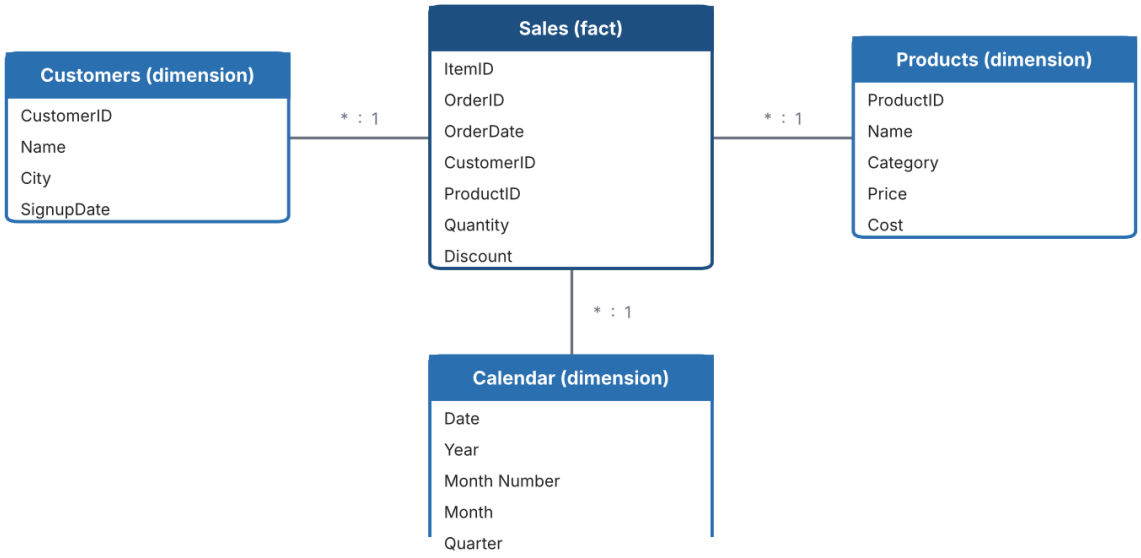
• The top 10 customers produce 71% of revenue, so losing one of them is a real risk. 8 registered customers have not ordered at all.

• Average profit per order line falls from Rs 15,541 with no discount to Rs 6,369 at a 20% discount.

Note: The dashboard previews in this report are drawn from the same sample data with Python to show the intended design and the correct numbers. Power BI's own visuals will look similar but not identical. The data is sample data created for this project; all amounts are in Rs.

1. Data model

Star schema: one fact table (Sales) with three dimension tables. Customers and Products filter Sales (one-to-many); Calendar is marked as the date table.



Relationships

Many side	One side	Direction
Sales[CustomerID]	Customers[CustomerID]	Single (one-to-many)
Sales[ProductID]	Products[ProductID]	Single (one-to-many)
Sales[OrderDate]	Calendar[Date]	Single (one-to-many)

After creating the Calendar table, right-click it and choose **Mark as date table**. Sort the Month column by Month Number so months appear in calendar order.

2. Measures (DAX)

Every number on the dashboard comes from a measure. The values below are the results for the full year, calculated from the same data. The DAX code is in Appendix B.

Measure	What it calculates	Value
Total Revenue	Sales after discount	Rs 22.58M
Total Cost	Purchase cost of items sold	Rs 17.16M
Total Profit	Revenue minus cost	Rs 5.42M
Profit Margin %	Profit divided by revenue	24.0%
Total Orders	Distinct order numbers	240
Active Customers	Customers with at least one order	32
Avg Order Value	Revenue per order	Rs 94,079
Units Sold	Total quantity	804
Discount Given	Money given away as discount	Rs 1.01M
Avg Profit per Order Line	Profit per row of the Sales table	Rs 14,123
Revenue Previous Month	Revenue of the month before (Dec 2025 view)	Rs 1.36M
MoM Growth %	Change vs previous month (Dec 2025 view)	+184.9%
Revenue YTD	Year-to-date revenue (Dec 2025 view)	Rs 22.58M
Revenue Share %	Share of all revenue (Laptop shown)	28.1%
Top 10 Customer Share %	Revenue share of the 10 biggest customers	71.0%

3. Dashboard design

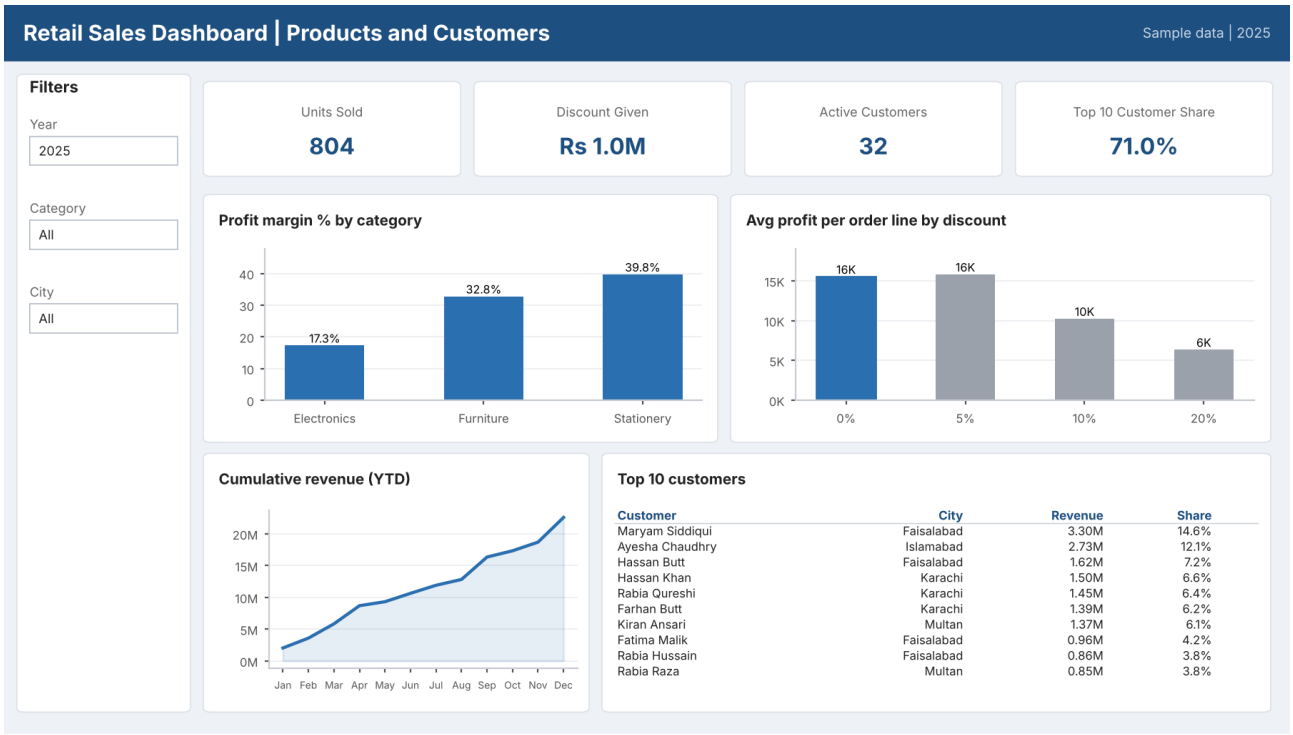
Page 1: Overview



Dashboard preview (design mockup with real values from the data).

Visual	Type	Fields
Card x5	Card	Total Revenue, Total Profit, Profit Margin %, Total Orders, Avg Order Value
Monthly revenue	Line chart	Axis: Calendar[Month] | Values: Total Revenue
Revenue by category	Column chart	Axis: Products[Category] | Values: Total Revenue
Top 5 products	Bar chart + Top N filter	Axis: Products[Name] | Values: Total Revenue | Top N = 5
Revenue by city	Column chart	Axis: Customers[City] | Values: Total Revenue
Filters	Slicers	Calendar[Year], Products[Category], Customers[City]

Page 2: Products and Customers



Dashboard preview (design mockup with real values from the data).

Visual	Type	Fields
Card x4	Card	Units Sold, Discount Given, Active Customers, Top 10 Customer Share %
Profit margin by category	Column chart	Axis: Products[Category] | Values: Profit Margin %
Avg profit by discount	Column chart	Axis: Sales[Discount] | Values: Avg Profit per Order Line
Cumulative revenue	Area chart	Axis: Calendar[Month] | Values: Revenue YTD
Top 10 customers	Table + Top N filter	Customers[Name], City, Total Revenue, Revenue Share %

Design guidelines

• Canvas 16:9 (1280 x 720), light grey page background, white tiles with a thin border.
• Colours: dark blue for headers, blue for the main series, grey for context, orange only to flag problems or highlights. Font: Segoe UI.
• KPI cards on top, trends in the middle, breakdowns at the bottom, filters on the left.
• Every visual has a clear title; no more than 6 to 7 visuals per page.

4. Insights and recommendations

What the data shows

• Revenue was Rs 22.58M with a profit margin of 24.0% across 240 orders. Average order value is Rs 94,079.

• November and December together bring 23% of the year's revenue; December alone grew +185% over November.

• Laptop is the biggest product (28% of revenue). Electronics is the top category with 58% of revenue.

• Stationery has the best margin (39.8%) but the smallest revenue; Electronics margin is 17.3%.

• Faisalabad is the strongest city (Rs 8.0M) and Lahore the weakest (Rs 659,778).

• The top 10 customers produce 71% of revenue, so losing one of them is a real risk. 8 registered customers have not ordered at all.

• Average profit per order line falls from Rs 15,541 with no discount to Rs 6,369 at a 20% discount.

Recommended actions

• Build stock and run campaigns before November; that is the peak season.

• Protect the top 10 customers with account management and loyalty offers.

• Cap discounts at 10% unless the extra volume is proven to pay for them.

• Send a first-purchase offer to the 8 customers who never ordered.

• Plan regional promotions for Lahore, the weakest city.

5. How to build it in Power BI Desktop

Create the folder `C:\PowerBI\RetailSales` and save these three CSV files in it:

File	Table	Columns	Rows
sales.csv	Sales	ItemID, OrderID, OrderDate, CustomerID, ProductID, Quantity, Discount	384
customers.csv	Customers	CustomerID, Name, City, SignupDate	40
products.csv	Products	ProductID, Name, Category, Price, Cost	15

1. Home > Get data > Text/CSV. Load the three files. (Or open Transform data > New Source > Blank Query > Advanced Editor and paste the Power Query code from Appendix B.)

2. In Power Query check the column types (dates as Date, amounts as Decimal) and click Close and Apply.

3. Modeling > New table. Paste the Calendar code from Appendix B. Mark it as the date table and sort Month by Month Number.

4. Open Model view and create the relationships from the table in section 1 (drag the key column from the Sales table to the dimension).

5. Right-click the Sales table > New measure. Paste each measure from Appendix B. Set formats (currency, percentage) in the Measure tools tab.

6. Report view > View > Page view > Actual size; set Format page > Canvas settings to 16:9 and a light grey background.

7. Add the visuals from the tables in section 3. Drag the fields exactly as listed; use Top N in the Filters pane where noted.

8. Add slicers and align everything to a grid (Format > Align). Add titles, turn off unnecessary gridlines and keep colours consistent.

9. Duplicate the page for page 2, replace the visuals, then File > Save. Export with File > Export > PDF for submission.

Appendix A: Sales list

First 45 records out of 384 rows of the Sales table (with names added from the dimension tables).

Orderid	Orderdate	Customer	City	Product	Quantity	Discount pct	Amount
1	2025-01-01	Maryam Siddiqui	Faisalabad	Office Chair	1	10.0	22,500
1	2025-01-01	Maryam Siddiqui	Faisalabad	Sofa Set	2	0.0	240,000
2	2025-01-03	Ayesha Chaudhry	Islamabad	Laptop	2	0.0	370,000
3	2025-01-08	Rabia Raza	Multan	File Folder	4	0.0	1,400
3	2025-01-08	Rabia Raza	Multan	Smartphone	2	5.0	180,500
4	2025-01-09	Kiran Ansari	Multan	Pen Set	4	0.0	1,800
5	2025-01-16	Hassan Malik	Karachi	Tablet	2	10.0	108,000
6	2025-01-17	Ayesha Chaudhry	Islamabad	Printer Paper	2	0.0	3,600
6	2025-01-17	Ayesha Chaudhry	Islamabad	Calculator	3	10.0	5,940
7	2025-01-18	Maryam Siddiqui	Faisalabad	Study Desk	2	0.0	70,000
7	2025-01-18	Maryam Siddiqui	Faisalabad	Sofa Set	2	10.0	216,000
8	2025-01-20	Maryam Siddiqui	Faisalabad	Bookshelf	3	5.0	51,300
9	2025-01-21	Hassan Sheikh	Faisalabad	Calculator	4	0.0	8,800
10	2025-01-23	Zainab Chaudhry	Karachi	Marker Set	4	20.0	2,880
11	2025-01-23	Noor Khan	Islamabad	Calculator	2	0.0	4,400
11	2025-01-23	Noor Khan	Islamabad	Study Desk	2	10.0	63,000
12	2025-01-23	Fatima Malik	Faisalabad	Smart Watch	2	0.0	44,000
12	2025-01-23	Fatima Malik	Faisalabad	Laptop	1	0.0	185,000
13	2025-01-26	Saad Chaudhry	Karachi	Smart Watch	2	20.0	35,200
13	2025-01-26	Saad Chaudhry	Karachi	Calculator	4	10.0	7,920
13	2025-01-26	Saad Chaudhry	Karachi	Smartphone	2	20.0	152,000
14	2025-01-27	Hassan Butt	Faisalabad	Smartphone	2	5.0	180,500
15	2025-01-28	Hira Malik	Multan	Headphones	2	0.0	17,000
15	2025-01-28	Hira Malik	Multan	Bookshelf	4	0.0	72,000
16	2025-02-04	Hassan Khan	Karachi	Printer Paper	2	0.0	3,600
16	2025-02-04	Hassan Khan	Karachi	Laptop	2	0.0	370,000
16	2025-02-04	Hassan Khan	Karachi	Bookshelf	1	0.0	18,000
17	2025-02-05	Hira Malik	Multan	Printer Paper	1	20.0	1,440
18	2025-02-05	Fatima Malik	Faisalabad	Laptop	1	20.0	148,000
18	2025-02-05	Fatima Malik	Faisalabad	File Folder	2	0.0	700
18	2025-02-05	Fatima Malik	Faisalabad	Printer Paper	1	0.0	1,800
19	2025-02-09	Ayesha Chaudhry	Islamabad	Notebook Pack	3	10.0	1,620
20	2025-02-10	Farhan Butt	Karachi	Office Chair	2	10.0	45,000
21	2025-02-11	Hassan Khan	Karachi	Pen Set	1	0.0	450
22	2025-02-12	Saad Malik	Islamabad	Smartphone	1	0.0	95,000
22	2025-02-12	Saad Malik	Islamabad	File Folder	4	0.0	1,400

Orderid	Orderdate	Customer	City	Product	Quantity	Discount pct	Amount
23	2025-02-12	Kiran Malik	Faisalabad	File Folder	3	20.0	840
24	2025-02-15	Kiran Malik	Faisalabad	Smartphone	2	0.0	190,000
25	2025-02-18	Zainab Qureshi	Multan	Bookshelf	3	0.0	54,000
26	2025-02-19	Hira Malik	Multan	File Folder	3	0.0	1,050
27	2025-02-19	Rabia Siddiqui	Karachi	Office Chair	2	0.0	50,000
28	2025-02-20	Ayesha Chaudhry	Islamabad	Headphones	4	5.0	32,300
28	2025-02-20	Ayesha Chaudhry	Islamabad	Calculator	3	5.0	6,270
29	2025-02-22	Maryam Siddiqui	Faisalabad	Sofa Set	1	0.0	120,000
30	2025-02-22	Hassan Butt	Faisalabad	Tablet	2	10.0	108,000

Appendix B: Power Query (M) and DAX code

Copy-paste ready. Part 1 loads the CSV files in Power Query; part 2 is DAX.

```
// ===== PART 1: POWER QUERY (M) - one query per table =====

// ----- Query: Sales -----
let
    Source = Csv.Document(
        File.Contents("C:\PowerBI\RetailSales\sales.csv"),
        [Delimiter = ",", Columns = 7, Encoding = 65001, QuoteStyle = QuoteStyle.Csv]),
    #"Promoted Headers" = Table.PromoteHeaders(Source, [PromoteAllScalars = true]),
    #"Changed Type" = Table.TransformColumnTypes(#"Promoted Headers", {
        {"ItemID", Int64.Type},
        {"OrderID", Int64.Type},
        {"OrderDate", type date},
        {"CustomerID", Int64.Type},
        {"ProductID", Int64.Type},
        {"Quantity", Int64.Type},
        {"Discount", type number}
    })
in
    #"Changed Type"

// ----- Query: Customers -----
let
    Source = Csv.Document(
        File.Contents("C:\PowerBI\RetailSales\customers.csv"),
        [Delimiter = ",", Columns = 4, Encoding = 65001, QuoteStyle = QuoteStyle.Csv]),
    #"Promoted Headers" = Table.PromoteHeaders(Source, [PromoteAllScalars = true]),
    #"Changed Type" = Table.TransformColumnTypes(#"Promoted Headers", {
        {"CustomerID", Int64.Type},
        {"Name", type text},
        {"City", type text},
        {"SignupDate", type date}
    })
in
    #"Changed Type"

// ----- Query: Products -----
let
    Source = Csv.Document(
        File.Contents("C:\PowerBI\RetailSales\products.csv"),
        [Delimiter = ",", Columns = 5, Encoding = 65001, QuoteStyle = QuoteStyle.Csv]),
    #"Promoted Headers" = Table.PromoteHeaders(Source, [PromoteAllScalars = true]),
    #"Changed Type" = Table.TransformColumnTypes(#"Promoted Headers", {
        {"ProductID", Int64.Type},
        {"Name", type text},
        {"Category", type text},
        {"Price", type number},
        {"Cost", type number}
    })
in
    #"Changed Type"

// ===== PART 2: DAX =====

-- ===== Calendar table (Modeling > New table) =====
Calendar =
ADDCOLUMNS (
    CALENDAR ( DATE ( 2025, 1, 1 ), DATE ( 2025, 12, 31 ) ),
    "Year", YEAR ( [Date] ),
    "Month Number", MONTH ( [Date] ),
    "Month", FORMAT ( [Date], "MMM" ),
    "Quarter", "Q" & ROUNDUP ( MONTH ( [Date] ) / 3, 0 )
)

-- ===== Measures (Modeling > New measure) =====
Total Revenue =
SUMX ( Sales, Sales[Quantity] * RELATED ( Products[Price] ) * ( 1 - Sales[Discount] ) )

Total Cost =
SUMX ( Sales, Sales[Quantity] * RELATED ( Products[Cost] ) )

Total Profit = [Total Revenue] - [Total Cost]

Profit Margin % = DIVIDE ( [Total Profit], [Total Revenue] )

Total Orders = DISTINCTCOUNT ( Sales[OrderID] )

Active Customers = DISTINCTCOUNT ( Sales[CustomerID] )

Avg Order Value = DIVIDE ( [Total Revenue], [Total Orders] )

Units Sold = SUM ( Sales[Quantity] )
Discount Given =
```

```
SUMX ( Sales, Sales[Quantity] * RELATED ( Products[Price] ) * Sales[Discount] )

Avg Profit per Order Line = DIVIDE ( [Total Profit], COUNTROWS ( Sales ) )

Revenue Previous Month =
CALCULATE ( [Total Revenue], DATEADD ( 'Calendar'[Date], -1, MONTH ) )

MoM Growth % =
DIVIDE ( [Total Revenue] - [Revenue Previous Month], [Revenue Previous Month] )

Revenue YTD = TOTALYTD ( [Total Revenue], 'Calendar'[Date] )

Revenue Share % =
DIVIDE ( [Total Revenue], CALCULATE ( [Total Revenue], ALL ( Products ) ) )

Top 10 Customer Share % =
VAR Top10 = TOPN ( 10, ALL ( Customers[Name] ), [Total Revenue], DESC )
RETURN
    DIVIDE (
        CALCULATE ( [Total Revenue], KEEPFILTERS ( Top10 ) ),
        CALCULATE ( [Total Revenue], ALL ( Customers ) )
    )
```